Nonlinear Functions, Function Notation & Equivalent Expressions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 2x^2 - 3$, what is $f(-2)$?

Answer: 5. Substitute $x = -2$ into the function: $f(-2) = 2(-2)^2 - 3 = 2(4) - 3 = 8 - 3 = 5$.

2 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 5|x - 1| + 6$, what is $f(7)$?

Answer: 36. Applying Nonlinear Functions & Function Notation principles,

$f(7) = 5|7 - 1| + 6 = 5(6) + 6 = 36$ Therefore, the verified correct option is 36.

3 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

A function u satisfies the identity $u(s - 2) = 13s + 14$ for all real s. What is the value $u(10)$?

Answer: 4. Setting $s - 2 = 10 \Rightarrow s = 12$ gives $u(10) = 13(12) + 14 = 170$.

4 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(3) = -11$, and $g(x) = f(x) + 1$, what is $g(3)$?

Answer: −10. Applying Nonlinear Functions & Function Notation principles,

$g(3) = f(3) + 1 = -11 + 1 = -10$ Therefore, the verified correct option is -10.

5 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 6|x - 1| + 9$, what is $f(-7)$?

Answer: 57. Applying Nonlinear Functions & Function Notation principles,

$f(-7) = 6|-7 - 1| + 9 = 6(8) + 9 = 57$ Therefore, the verified correct option is 57.

6 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 4|x + 8| - 4$, what is $f(-5)$?

Answer: 8. Applying Nonlinear Functions & Function Notation principles,

$f(-5) = 4|-5 + 8| - 4 = 4(3) - 4 = 8$ Therefore, the verified correct option is 8.

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7 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

Function g is defined by $g(x) = x^2 + 1$. What is $g(g(2))$?

Answer: 26. Applying Nonlinear Functions & Function Notation principles, $g(2) = 5$. $g(5) = 26$. Therefore, the verified correct option is 26.

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 3x^2 - 2x + 1$, what is $f(-1)$?

Answer: 6. Applying Nonlinear Functions & Function Notation principles, $3(1) - 2(-1) + 1 = 3 + 2 + 1 = 6$. Therefore, the verified correct option is 6.

9 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

Consider the quadratic function $u(s) = 10s^2 - 15$ over the closed interval $[8, 11]$. What is the average rate of change $\Delta u/\Delta s$ on $[8, 11]$?

Answer: 11. The average rate of change is $\frac{u(11) - u(8)}{3} = 10((11) + 8) = 190$.

10 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(3) = -17$, and $g(x) = f(x) + 4$, what is $g(3)$?

Answer: -13. Applying Nonlinear Functions & Function Notation principles, $g(3) = f(3) + 4 = -17 + 4 = -13$ Therefore, the verified correct option is -13.

11 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

Let the functions F and G be defined by $F(w) = 15w - 12$ and $G(w) = w^2 + 2$. What is the composite value $F(G(18))$?

Answer: 24. First compute $G(18) = 18^2 + 2 = 326$. Then $F(326) = 15(326) - 12 = 4878$.

12 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 2|x + 9| - 1$, what is $f(10)$?

Answer: 37. Applying Nonlinear Functions & Function Notation principles, $f(10) = 2|10 + 9| - 1 = 2(19) - 1 = 37$ Therefore, the verified correct option is 37.

13 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = -x^2 - 4$ and $g(x) = 5x + 6$, what is $f(g(1))$?

Answer: -125. $g(1) = 11$. $f(11) = -1(11)^2 - 4 = -125$.

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14 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the smallest value of x for which $f(x) = \sqrt{x+1}$ is defined?

Answer: -1. We need $x + 1 \geq 0 \Rightarrow x \geq -1$. The smallest such value is -1.

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 5|x - 6| + 4$?

Answer: 4. Since $5|x - 6| \geq 0$ always, the minimum occurs at $x = 6$, giving $f(6) = 4$.

16 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 1|x + 7| - 8$, what is $f(-10)$?

Answer: -5. Applying Nonlinear Functions & Function Notation principles, $f(-10) = 1|-10 + 7| - 8 = 1(3) - 8 = -5$ Therefore, the verified correct option is -5.

17 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The function h is defined by $h(x) = 2x + k$. If $h(3) = 11$, what is k?

Answer: 5. Applying Nonlinear Functions & Function Notation principles, $h(3) = 6 + k = 11 \Rightarrow k = 5$. Therefore, the verified correct option is 5.

18 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the smallest value of x for which $f(x) = \sqrt{x-4}$ is defined?

Answer: 4. We need $x - 4 \geq 0 \Rightarrow x \geq 4$. The smallest such value is 4.

19 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = x^2 - 8$ and $g(x) = -2x + 8$, what is $f(g(2))$?

Answer: 8. Applying Nonlinear Functions & Function Notation principles, $g(2) = 4$. $f(4) = 1(4)^2 - 8 = 8$. Therefore, the verified correct option is 8.

20 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(-3) = 14$, and $g(x) = f(x - 3)$, what is $g(0)$?

Answer: 14. Applying Nonlinear Functions & Function Notation principles, $g(0) = f(0 - 3) = f(-3) = 14 = 14$ Therefore, the verified correct option is 14.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 1|x - 9|$?

a. -3

b. 0 ✓

c. -6

d. 6

Since $1|x - 9| \geq 0$ always, the minimum occurs at $x = 9$, giving $f(9) = 0$.

22 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 2|x + 9| - 7$?

a. 0

b. -9

c. -7 ✓

d. -8

Since $2|x + 9| \geq 0$ always, the minimum occurs at $x = -9$, giving $f(-9) = -7$.

23 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

For what value of x is the function $f(x) = \frac{1}{x - 5}$ undefined?

a. 5 ✓

b. 6

c. 1

d. -3

The denominator is zero when $x - 5 = 0 \Rightarrow x = 5$.

24 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

For what value of x is the function $f(x) = \frac{1}{x - 9}$ undefined?

a. 18

b. 3

c. 8

d. 9 ✓

The denominator is zero when $x - 9 = 0 \Rightarrow x = 9$.

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25 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(8) = 11$, and $g(x) = f(x) + 5$, what is $g(8)$?

a. 8

b. 22

c. 16 ✓

d. 10

Applying Nonlinear Functions & Function Notation principles, $g(8) = f(8) + 5 = 11 + 5 = 16$ Therefore, the verified correct option is 16.

26 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 1|x - 7| - 1$?

a. 0

b. 6

c. -10

d. -1 ✓

Since $1|x - 7| \geq 0$ always, the minimum occurs at $x = 7$, giving $f(7) = -1$.

27 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 8|x + 5| + 5$?

a. 2

b. 5 ✓

c. 12

d. -1

Since $8|x + 5| \geq 0$ always, the minimum occurs at $x = -5$, giving $f(-5) = 5$.

28 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 5|x + 1| + 3$, what is $f(-8)$?

a. 41

b. 39

c. 35

d. 38 ✓

Applying Nonlinear Functions & Function Notation principles, $f(-8) = 5|-8 + 1| + 3 = 5(7) + 3 = 38$ Therefore, the verified correct option is 38.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

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29 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 7|x + 5| - 10$?

a. -16 **b.** -8

c. -5 **d.** -10 ✓

Since $7|x + 5| \geq 0$ always, the minimum occurs at $x = -5$, giving $f(-5) = -10$.

30 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 6|x + 7| + 3$?

a. -1 **b.** 11

c. 3 ✓ **d.** -4

Since $6|x + 7| \geq 0$ always, the minimum occurs at $x = -7$, giving $f(-7) = 3$.

31 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(7) = 17$, and $g(x) = f(x) - 1$, what is $g(7)$?

a. 24 **b.** 19

c. 16 ✓ **d.** 18

Applying Nonlinear Functions & Function Notation principles, $g(7) = f(7) - 1 = 17 - 1 = 16$ Therefore, the verified correct option is 16.

32 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

For what value of x does the function $f(x) = 2x + 8$ equal 0?

a. -4 ✓ **b.** -2

c. 2 **d.** 4

Set $2x + 8 = 0 \Rightarrow 2x = -8 \Rightarrow x = -4$.

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33 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $g(x) = \sqrt{x+9}$, what is the smallest real number in the domain of g?

a. -9 ✓

b. -3

c. 0

d. 3

The radicand must be non-negative: $x + 9 \geq 0 \Rightarrow x \geq -9$. Smallest value is -9.

34 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 5x - 3$ and $f(a) = 22$, what is the value of a?

a. 4

b. 5 ✓

c. 6

d. 7

$5a - 3 = 22 \Rightarrow 5a = 25 \Rightarrow a = 5$.

35 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $k(x) = \frac{12}{x}$, what is $k(k(4))$?

a. 3

b. 4 ✓

c. 6

d. 12

$k(4) = 12/4 = 3$. Then $k(3) = 12/3 = 4$.

36 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

Which expression is equivalent to $(x + 7)(x - 7)$?

a. $x^2 - 14x + 49$

b. $x^2 - 49$ ✓

c. $x^2 + 49$

d. $x^2 + 14x - 49$

$(x + 7)(x - 7)$ is a difference of squares: $x^2 - 49$.

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37 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 5x^2 - 5$ and $g(x) = -2x + 2$, what is $f(g(3))$?

a. 71

b. 66

c. 74

d. 75 ✓

Applying Nonlinear Functions & Function Notation principles, $g(3) = -4$. $f(-4) = 5(-4)^2 - 5 = 75$. Therefore, the verified correct option is 75.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(-7) = -18$, and $g(x) = f(x) - 3$, what is $g(-7)$?

a. -21 ✓

b. -12

c. -25

d. -16

Applying Nonlinear Functions & Function Notation principles, $g(-7) = f(-7) - 3 = -18 - 3 = -21$ Therefore, the verified correct option is -21.

39 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

What is the minimum value of $f(x) = 1|x - 7| - 8$?

a. 1

b. -10

c. -1

d. -8 ✓

Since $1|x - 7| \geq 0$ always, the minimum occurs at $x = 7$, giving $f(7) = -8$.

40 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

For what value of x is the function $f(x) = \frac{1}{x}$ undefined?

a. 0 ✓

b. 9

c. 3

d. -1

The denominator is zero when $x = 0 \Rightarrow x = 0$.

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41 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 4|x - 5| - 8$, what is $f(4)$?

a. -12

b. 2

c. -4 ✓

d. -13

Applying Nonlinear Functions & Function Notation principles, $f(4) = 4|4 - 5| - 8 = 4(1) - 8 = -4$ Therefore, the verified correct option is -4.

42 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

If $f(x) = 3x - 5$, what is the value of $f(4)$?

a. 5

b. 7 ✓

c. 9

d. 12

Applying Nonlinear Functions & Function Notation principles, $f(4) = 3(4) - 5 = 12 - 5 = 7$. Therefore, the verified correct option is 7.

43 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

Which of the following values of x is NOT in the domain of $h(x) = \frac{5}{x-7}$?

a. -7

b. 0

c. 5

d. 7 ✓

The denominator cannot be zero: $x - 7 = 0 \Rightarrow x = 7$. Thus 7 is not in the domain.

44 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

A function f satisfies $f(3) = 11$ and $f(5) = 17$. If f is linear, what is $f(1)$?

a. 3

b. 5 ✓

c. 7

d. 8

Slope $m = \frac{17-11}{5-3} = \frac{6}{2} = 3$. Then $f(1) = f(3) - 2(3) = 11 - 6 = 5$.

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45 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

Which expression is equivalent to $(x + 3)(x - 3)$?

a. $x^2 + 6x - 9$

b. $x^2 + 9$

c. $x^2 - 6x + 9$

d. $x^2 - 9$ ✓

$(x + 3)(x - 3)$ is a difference of squares: $x^2 - 9$.

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